



Future interns –cyber security

Phishing Email Detection & Awareness System

Detecting deceptive emails with machine learning, and building organizational resilience through user awareness.

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AGENDA

What We Will Cover



The Phishing Problem

Scope, statistics, and why it matters today



Objectives & Scope

What the system aims to achieve



System Architecture

How detection and awareness modules interact



Detection Methodology

Feature engineering and ML pipeline



Results & Evaluation

Model performance and comparison



Awareness Module

Simulated phishing training for users



Challenges & Future Scope

Limitations and roadmap ahead

Phishing Is Still the #1 Entry Point for Cyberattacks

Attackers no longer need to “hack” a system when they can simply trick a person into opening the door. Email remains the most exploited channel for credential theft, malware delivery, and business fraud.

91%

of cyberattacks begin with a
phishing email

\$4.9M

average cost of a data breach
caused by phishing

1 in 4

employees click a simulated
phishing link

3.4B

phishing emails sent worldwide
every single day

Sources: Verizon DBIR, IBM Cost of a Data Breach, industry phishing simulation benchmarks (representative figures).

Understanding the Attack: Types of Phishing



Deceptive Phishing

Mass emails impersonating trusted brands to harvest credentials.



Spear Phishing

Highly targeted messages using personal or organizational context.



Whaling

Attacks aimed at executives to authorize fraudulent transactions.



Clone Phishing

A legitimate email is copied and its links/attachments swapped.



Smishing / Vishing

Phishing delivered via SMS or voice calls instead of email.



Pharming

Redirecting users from legitimate sites to fraudulent lookalikes.

Project Objectives

-  Build an ML-based classifier that detects phishing emails with high precision and low false positives.
-  Engineer meaningful features from headers, URLs, and message content that reveal deceptive intent.
-  Design a lightweight awareness module that trains users through simulated phishing campaigns.
-  Deliver actionable alerts and reports so security teams can respond quickly.
-  Measure impact through detection accuracy and improvement in human reporting behavior.

Existing Approaches & Their Gaps

Approach	Strength	Limitation
Blocklist / Signature Filters	Fast, low compute	Fails on new (zero-day) phishing URLs
Rule-Based Heuristics	Interpretable, easy to deploy	Easily bypassed by minor wording changes
Classical ML (Naive Bayes, SVM)	Good accuracy on known patterns	Needs manual feature tuning, drifts over time
Deep Learning (LSTM, BERT-based)	Captures semantic & contextual cues	Heavy, needs large labeled datasets

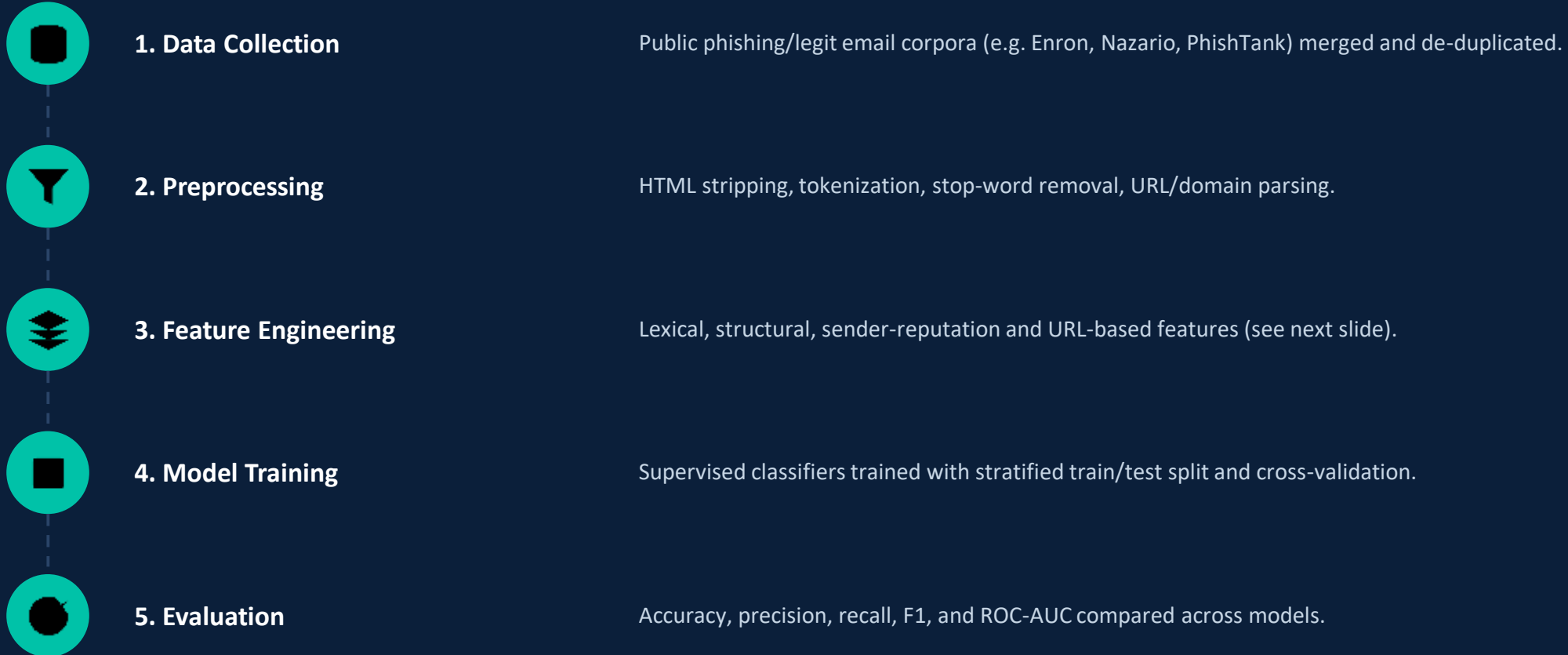
Our system combines engineered features with a supervised model, and pairs detection with a human-awareness loop that most existing tools ignore.

End-to-End System Architecture



The pipeline runs continuously: every processed email — flagged or not — strengthens the model and the awareness dataset.

Detection Methodology



What the Model Looks At



URL-Based

- IP address in URL
- URL shortening services
- @ symbol / redirects
- Domain age & mismatch



Content-Based

- Urgency & fear keywords
- Spelling/grammar errors
- Generic greetings
- Suspicious attachments



Sender-Based

- SPF / DKIM / DMARC failures
- Reply-to ≠ From address
- Newly registered domains
- Spoofed display names

Tools & Technologies



Python

Core language for data processing & modeling



Scikit-learn

Classical ML models & evaluation metrics



Pandas / NumPy

Data wrangling and feature engineering



PhishTank / Enron

Public labeled datasets for training



Flask API

Serves the trained model for real-time scoring



Email Connector

IMAP/SMTP hook to scan live inboxes



Matplotlib / Seaborn

Visualizing model performance

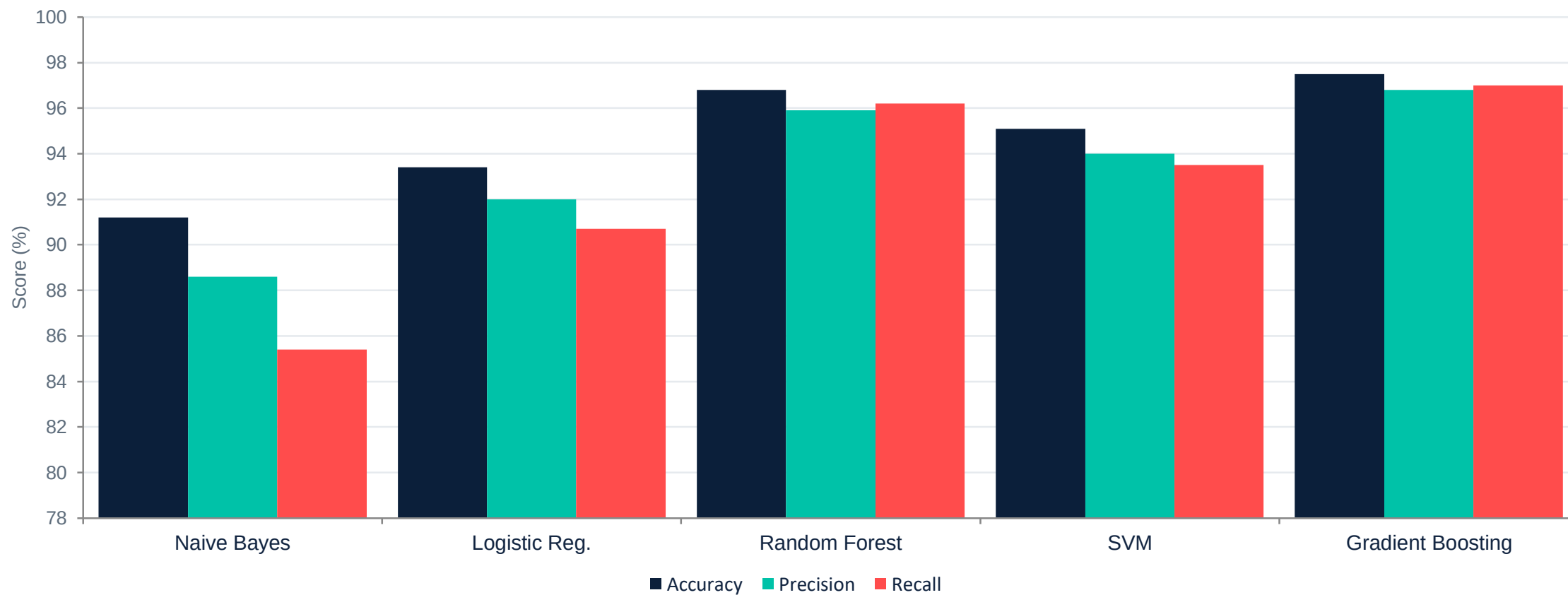


Awareness Module

Simulated campaign + training content engine

Model Comparison

Accuracy, precision and recall across candidate classifiers on the held-out test set.



RESULTS

Best Model: Gradient Boosting

Confusion Matrix (test set, n = 2000)

963 True Legit	37 False Phishing
22 False Legit	978 True Phishing

Predicted →

97.5%

Accuracy

96.8%

Precision

97.0%

Recall

96.9%

F1-Score

ROC-AUC: 0.991 — strong separation between phishing and legitimate email classes across nearly all thresholds.

The Awareness Module

Technology alone can't stop every attack — so the system also trains the people behind the inbox.



Simulated Campaigns

Realistic mock-phishing emails sent periodically to staff.



Behavior Tracking

Logs who clicked, who reported, and response time.



Just-in-Time Training

Instant micro-lesson shown to anyone who clicks a simulated link.



Awareness Score

Individual & team risk scores tracked over time to show improvement.

What the Analyst Sees



HIGH RISK — Phishing Probability: 94%

Subject: "Urgent: Verify Your Account Within 24 Hours"

Sender: security-alert@paypa1-secure.com (SPF: fail)

Flags: Lookalike domain · Urgency language · Shortened URL

Recommended Action: Quarantine and notify recipient

Quarantine

Mark Safe



Explainable, Not a Black Box

Every flagged email shows the exact signals that triggered it — domain mismatch, urgency language, failed authentication — so analysts trust and can override the model.



Real-Time Scoring

Emails are scored in under 200ms at the gateway, so risky mail is quarantined before a user ever opens it.

Challenges Faced During Development



Imbalanced & Noisy Data

Public datasets skew toward phishing samples; required careful resampling and cleaning.



Evasive New Attack Patterns

Attackers constantly change wording and domains — features need regular retraining.



False Positives

Overly aggressive flagging risks legitimate mail being blocked and user trust lost.



User Engagement

Getting staff to actually complete awareness training required gamified nudges.

Where This Project Goes Next



Transformer-Based Models

Fine-tune BERT/DistilBERT for deeper contextual understanding of email text.



Browser & Mobile Extension

Real-time warnings directly inside webmail and mobile mail clients.



Multilingual Detection

Extend feature set and training data beyond English-language phishing.



Adaptive Learning

Continuous retraining pipeline using analyst feedback as new labels.

Building a Safer Inbox — One Layer at a Time

- A supervised ML pipeline detects phishing emails with 97%+ accuracy using engineered content, URL and sender features.
- An awareness module closes the human gap by training the people who ultimately decide whether to click.
- Together, technology and training form a defense-in-depth approach — far stronger than either alone.

“The strongest firewall is a well-trained inbox.”

ABOUT ME

Let's Connect

Aspiring cybersecurity professional passionate about applying machine learning to real-world security problems.



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Thank You

Questions & Discussion Welcome